

New GAS and Moderations reports

14 May 2026 - Prof Arnold J Rix

Agenda

- New GAs to be implemented
 - Important for Project 448 - GA 1-6, 8 & 9
- New Moderation reports
- A unified approach to exemption for repeaters

New GAs - Project 448 Examination

- Important for Project 448 - GA 1-6, 8 & 9
- The **skripsie template will be adjusted** to specifically cater for the changes in the GAs and allow the students to think about these and incorporate it accordingly.
- This will assist in showing a **paper trail for moderation** purposes.
- Be aware of this and motivate your students to spend time and complete the sections sensibly.
- **Lecturers to provide feedback** if you already introduce and/or cover some of these new topics in your module or if you think you could do so. This will assist in preparing students to meet the new GAs.

New GAs - Complex Engineering Problems

Complex engineering problems

(a) These require a fundamentals-based, first principles analytical approach, use in-depth engineering knowledge and have one or more of the following characteristics:

- i. A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- ii. Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much of which is at the forefront of the discipline.
- iii. Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- iv. Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- v. Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

(b) and have some or all of the following characteristics:

- i. Involve wide-ranging and/or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements.
- ii. Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models.
- iii. Involve infrequently encountered issues or novel problems.
- iv. Address problems not encompassed by standards and codes of practice for professional engineering.
- v. Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs.
- vi. Address high-level problems with many components or sub-problems that may require a systems approach.

Key Differences - Complex Engineering Problems

- **Shift from Traditional Technical Focus → Modern Systems-Based Engineering**
- **Stronger first-principles and theory-based approach**
Explicit focus on analytical fundamentals and specialist knowledge at the forefront of the discipline.
- **Introduction of sustainability and lifecycle thinking**
Includes resource efficiency, environmental impact, whole-life cost, re-use of resources, and net-zero carbon considerations.
- **Greater emphasis on research and innovation**
Requires engagement with current research literature, critical thinking, creativity, and emerging issues.
- **Expansion beyond purely technical problems**
Incorporates ethical, legal, political, economic, societal, and future-focused considerations.
- **Increased collaboration and stakeholder engagement**
Highlights interdisciplinary teamwork and working with diverse stakeholder groups.
- **Explicit systems engineering perspective**
Greater focus on complex systems, integration, and systems-level problem solving.

Overall Shift

The new version broadens engineering competence from mainly technical problem-solving to sustainable, research-informed, multidisciplinary, and systems-oriented engineering practice.

New GAs - GA 1: Problem Solving

Identify, formulate, **research literature** and analyse complex engineering problems **reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.**

Associated knowledge and attitude profile:

- A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline, much of which is at the forefront of the discipline.

Key Differences - GA 1: Problem Solving

- **Research-informed problem solving**
Adds engagement with research literature and substantiated conclusions.
- **Stronger first-principles and theory-based approach**
Greater emphasis on mathematics, natural sciences, and engineering fundamentals.
- **Sustainability and holistic considerations**
Introduces sustainable development and broader societal awareness into engineering decisions.
- **Expanded analytical and computational requirements**
Includes data analysis, statistics, modelling, and computer/information science tools.
- **Higher expectation of specialist knowledge**
Requires advanced engineering knowledge aligned with current and emerging practice areas.

Overall Shift

GA1 moves from traditional engineering problem solving toward a more **research-driven, sustainability-focused**, data-intensive, and theory-based engineering competency.

New GAs - GA 2: Application of scientific and engineering knowledge

Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.

Associated knowledge and attitude profile:

- A systematic, theory-based understanding of the natural sciences applicable to the discipline **and awareness of relevant social sciences.**
- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline, much of which is at the forefront of the discipline.

Range statement:

Mathematics, natural science and engineering sciences are applied in formal analysis and modelling of engineering situations, and for reasoning about and conceptualising engineering problems.

New GAs - GA 3: Engineering design

Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.

Associated knowledge and attitude profile:

Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon and similar concepts, that supports engineering design and operations in a practice area.

Range statement:

Design problems used in exit-level assessment must conform to the definition of a complex engineering problem. A major design problem should be used to provide evidence. The design knowledge base and components, systems, engineering works, products or processes to be designed are dependent on the discipline or practice area. **Appropriate consideration must be given to public health and safety, whole-life cost and net zero carbon, as well as resource, cultural, societal and environmental considerations, as required.**

Key Differences - GA 3: Engineering design

- **Greater emphasis on creative solution development**
Focus shifts from performing design to designing creative solutions for complex engineering problems.
- **Stronger sustainability and lifecycle considerations**
Introduces efficient resource use, environmental impacts, whole-life cost, re-use of resources, and net-zero carbon considerations.
- **Broader societal and public impact awareness**
Explicit inclusion of public health and safety, cultural, societal, and environmental considerations.
- **More holistic engineering design approach**
Design now extends beyond technical functionality to include operational, sustainability, and societal requirements.

Overall Shift

GA3 moves from a primarily technical design competency toward a more **creative, sustainability-driven, socially aware, and holistic engineering design approach.**

New GAs - GA 4: Investigations, experiments and data analysis

Demonstrate competence to conduct investigations of **complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.**

Associated knowledge and attitude profile:

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

Range statement:

The balance of investigation and experiment should be appropriate to the discipline. Research methodology is to be applied in research or an investigation where the student engages with selected knowledge in the research literature of the discipline.

Note: An investigation differs from a design in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact can be produced.

Key Differences - GA 4: Investigations, experiments and data analysis

- **Investigations explicitly linked to complex engineering problems**
Stronger emphasis on applying investigations within complex engineering contexts.
- **Formal research methods and research-based knowledge introduced**
Adds research methodology, design of experiments, and synthesis of information.
- **Greater focus on data analysis and substantiated conclusions**
Expands from conducting experiments to analysing, interpreting, and validating conclusions from data.
- **Critical thinking and creativity explicitly included**
Requires engagement with current research literature, critical thinking, and creative approaches to emerging issues.

Overall Shift

GA4 evolves from traditional experimental competency toward a more **research-intensive, analytical, evidence-based, and critically driven investigation approach.**

New GAs - GA 5: Use of engineering tools

Demonstrate competence to **create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.**

Range statement:

A range of **techniques, resources and modern engineering and IT tools** appropriate to the disciplinary designation of the programme.

Key Differences - GA 5: Use of engineering tools

- **Shift from using tools to creating, selecting, and applying tools**
Greater emphasis on engineering judgement and tool selection.
- **Explicit recognition of tool limitations**
Engineers must now understand the constraints and limitations of engineering and IT tools.
- **Stronger focus on modern engineering and IT tools**
Introduces modern digital tools, prediction, and modelling capabilities.
- **Broader emphasis on techniques and resources**
Expands beyond software and infrastructure to include techniques, resources, and integrated engineering tools.

Overall Shift

GA5 evolves from basic competency in using engineering tools toward a more **advanced, critical, and digitally integrated capability** focused on modern engineering methods, modelling, and informed tool application.

New GAs - GA 6: Professional and technical communication

Demonstrate competence to communicate effectively **and inclusively on complex engineering activities**, both orally and in writing, with the engineering community **and society** at large, **taking into account cultural, language, and learning differences**.

Range statement:

Material to be communicated is in an academic or simulated professional context. Audiences range from engineering peers, management and lay persons, using appropriate academic or professional discourse. Written reports range from short (300 to 1 000 words, plus tables and diagrams) to long (10 000 to 15 000 words plus tables, diagrams and appendices), covering material at exit-level. Methods of providing information include the conventional methods of the discipline, for example engineering drawings and design documentation, as well as subject-specific methods.

Key Differences - GA 6: Professional and technical communication

- Introduction of **inclusive communication**
Communication must now be effective and inclusive across diverse audiences.
- **Broader societal focus**
Expands from engineering audiences to include society at large.
- **Recognition of diversity considerations**
Explicit consideration of cultural, language, and learning differences.
- **Greater emphasis on communication within complex engineering activities**
Communication is more directly linked to complex engineering contexts and multidisciplinary engagement.

Overall Shift

GA6 evolves from traditional professional communication toward a more **inclusive, socially aware, and diversity-sensitive communication** competency suited to complex engineering environments.

New GAs - GA 7: The engineer and the world

Demonstrate critical awareness of the sustainable development impacts on society, the economy, sustainability, health and safety, legal frameworks and the environment.

Associated knowledge and attitude profile:

- A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon and similar concepts, that supports engineering design and operations in a practice area.
- Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

Range statement:

The combination of social, workplace (industrial) and physical environmental factors must be appropriate to the discipline or other designation of the qualification. Comprehension of the role of engineering in the world and identified issues in engineering practice in the discipline: health, safety and environmental protection, risk assessment and management, and the impacts of engineering activity: economic, social, cultural, environmental and sustainability.

Developmental considerations:

- Ability to self-reflect and show contextual awareness of social, workplace and governmental environments through exposure to complex, multi-disciplinary and/or unfamiliar problems.
- Ability to identify and position a design/artefact in the bigger picture and use appropriate judgement (intentionally incorporate multiple perspectives) to obtain a final solution or product.
- Ability to listen and interpret information from a variety of stakeholders to appropriately position identified problems/challenges/opportunities in the relevant context.

Key Differences - GA 7: The engineer and the world

- **Expanded sustainability focus**
Broadens from environmental and industrial impact to include the economy, legal frameworks, public health, safety, and sustainable development.
- **Greater emphasis on engineering's role in society**
Explicit focus on professional responsibility, public safety, and sustainable development.
- **Introduction of systems and contextual awareness**
Requires self-reflection, contextual awareness, and understanding of complex multidisciplinary environments.
- **Stronger stakeholder and societal engagement focus**
Adds listening to and interpreting information from diverse stakeholders to position engineering solutions appropriately.
- **Broader lifecycle and sustainability considerations**
Includes efficient resource use, whole-life cost, re-use of resources, and net-zero carbon concepts.

Overall Shift

GA7 evolves from general sustainability awareness toward a more **holistic, socially responsive, stakeholder-aware, and sustainability-driven understanding of engineering's role in the world.**

New GAs - GA 8: Individual and collaborative teamwork

Demonstrate competence to function effectively as an individual **and as a member or leader in diverse and inclusive teams**, and in multi-disciplinary, **face-to-face, remote and distributed settings**.

Associated knowledge and attitude profile:

Knowledge of professional ethics, responsibilities and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Range statement:

Multi-disciplinary tasks require co-operation across at least one disciplinary boundary. Co-operating disciplines may be engineering disciplines with different fundamental bases other than that of the programme or may be outside engineering.

Developmental considerations:

- **Develop the ability to self-reflect and self-evaluate within an interpersonal engagement towards enabling appropriate understanding of self and other team members.**
- **Develop the ability to listen and interpret different motivations, personalities or workstyles within a team context towards enabling functional team dynamics.**
- **Knowledge of team cohesion and dynamics, motivational styles, frameworks for conflict and tension resolution and the ability to apply these.**
- **Ability to negotiate and manage time and project components related to interpersonal needs and agendas. Time management also includes understanding the value of time, and determining if a task is better (cheaper) achieved by a single person or a team.**

Key Differences - GA 8: Individual and collaborative teamwork

- **Greater emphasis on inclusive and diverse teamwork**
Explicit focus on diversity, inclusion, mutual respect, and awareness of different backgrounds and abilities.
- **Expanded teamwork environments**
Adds remote, distributed, and face-to-face collaboration settings.
- **Leadership and interpersonal skills introduced**
Includes functioning as both a team member and leader, with stronger emphasis on interpersonal engagement.
- **Stronger focus on team dynamics and conflict management**
Introduces self-reflection, communication, negotiation, conflict resolution, and understanding team dynamics.
- **Project and time management considerations added**
Includes managing time, interpersonal needs, and project coordination within teams.

Overall Shift

GA8 evolves from basic multidisciplinary teamwork toward a more **inclusive, leadership-oriented, interpersonal, and collaborative teamwork competency** suited to modern engineering environments.

New GAs - GA 9: Independent learning ability

Demonstrate competence to engage in independent learning through well-developed learning skills.

Associated knowledge and attitude profile:

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

Range statement:

Operate independently in complex contexts recognising the need for and having the preparation and ability for

- i) independent and life-long learning,
- ii) adaptability to new and emerging technologies, and
- iii) critical thinking in the broadest context of technological change.

Developmental considerations:

- Openness to constructive feedback, awareness of own limitations, ability to cope with the discomfort of uncertainty and having access to a range of approaches, reflective self-evaluation, curiosity and proactive engagement, resilience, confidence to ask for help and draw from a broad range of stakeholders.
- Reflection of self-learning to begin to recognise if what has been covered meets the needs of the activity or task.

Key Differences - GA 9: Independent learning ability

- Stronger emphasis on **lifelong learning**
Explicit recognition of continuous, independent, and lifelong learning.
- Greater focus on **adaptability and emerging technologies**
Adds preparedness for technological change and evolving engineering environments.
- Critical thinking and research engagement introduced
Includes engagement with research literature, critical thinking, and creative evaluation of emerging issues.
- Expanded self-awareness and personal development focus
Introduces reflective self-evaluation, resilience, openness to feedback, and awareness of personal limitations.
- Greater emphasis on proactive and collaborative learning behaviour
Encourages curiosity, seeking help, stakeholder engagement, and coping with uncertainty.

Overall Shift

GA9 evolves from basic independent learning toward a more **lifelong, adaptive, reflective, research-informed, and resilient learning competency** suited to rapidly changing engineering environments.

New GAs - GA 10: Engineering professionalism

Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws.

Associated knowledge and attitude profile:

Knowledge of professional ethics, responsibilities and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Range statement:

Evidence includes case studies typical of engineering practice situations in which the graduate is likely to participate. **An understanding of the need for diversity and inclusion is required.**

Developmental considerations:

- **Self-management, professional responsibility and awareness of expertise and limitations, good judgement, process of on-going self-reflection and evaluation.**
- **Timeous, clear, realistic communication of risks and concerns and feedback on progress.**
- **Self-efficacy, accepting feedback and consequences and commitment.**

Key Differences - GA 10: Engineering professionalism

- Stronger emphasis on **ethical application and legal compliance**
Explicit requirement to apply ethical principles and adhere to national and international laws.
- Greater focus on **diversity and inclusion**
Introduces awareness of diversity, inclusion, mutual respect, and inclusive professional attitudes.
- Expanded professional responsibility and self-management expectations
Includes self-reflection, awareness of limitations, accountability, and professional judgement.
- Greater emphasis on communication and feedback
Adds timely communication of risks, concerns, progress, and acceptance of feedback and consequences.
- Broader professional development focus
Includes self-efficacy, commitment, ongoing self-evaluation, and professional growth.

Overall Shift

GA10 evolves from general professional and ethical awareness toward a more **accountable, inclusive, legally aware, self-reflective, and professionally engaged engineering practice competency.**

New GAs - GA 11: Project management and finance

Demonstrate knowledge and understanding of engineering management principles and economic decision-making.

Range statement:

Basic techniques from economics and project management applied to one's own work, as a member and leader in a team, and to manage projects in multi-disciplinary environments.

Thank you
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